TrustStack

A Local-First, Evidence-Grounded Evaluation Standard for Trustworthy AI

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1. Introduction and Problem Setting

TrustStack starts from a simple problem: a fluent answer is not enough when a user must decide whether an AI output is safe, supported, and auditable before using it.

Research question

How can a user judge whether an AI answer is trustworthy when conventional chat interfaces expose fluency but hide evidence quality, contradiction risk, and review conditions?

Project thesis

TrustStack reframes trust as an evidence-grounded evaluation task. It turns trust into a structured review artifact built from evidence, scoring, risk signals, and explanation.

Core task

Grounded trust review

Primary risk

Unsupported confidence

Decision basis

Evidence + diagnostics

2. Related Work and Contribution

Most AI systems focus on generating answers. TrustStack focuses on checking whether those answers deserve trust.

RAG systems

Pull evidence before answering.

Reliability research

Good sounding text can still be wrong.

Evaluation frameworks

Teams need repeatable scoring and reports.

TrustStack contribution

One local-first workflow that checks evidence, scores trust, and produces report-ready outputs.

3. Data

TrustStack evaluates AI with both real user evidence and benchmark data.

Live evidence path

- PDF, DOCX, TXT, and Markdown uploads
- Split into chunks and stored in MongoDB
- Used for grounded question answering

Benchmark path

- Seven synthetic stress tests
- Public slices from SciFact and HotpotQA
- Same scoring logic across all cases

User uploads | Synthetic stress tests | External benchmarks

4. Methods and System Architecture

The system follows a simple five-step pipeline so users can see where trust is gained or lost.

Pipeline

1. Ingest
2. Retrieve
3. Generate
4. Evaluate
5. Review / Export

Implementation

- MongoDB stores documents, chunks, and run history
- Retrieval uses vector search or lexical fallback
- Answers remain grounded with deterministic fallback

Verification stack

- Backend unit tests
- Real Mongo integration tests
- Browser E2E tests
- Served-stack smoke verification

5. TrustStack Evaluation Standard v2.0

TrustStack turns one answer into a full review record instead of a single hidden score.

Ten scored dimensions

Retrieval, support, citation traceability, claims, contradiction, completeness, abstention, discipline, safety, and calibration.

Verdicts

Pass / Review / Fail

Diagnostics

Claims + checks + flags

Output

Reproducible report artifact

Why this matters

The user can see why confidence changed, what support was missing, and when review is required.

6. Synthetic Benchmark Findings

Synthetic tests answer one question: does the score move the right way when evidence quality changes?

What was stressed

aligned evidence, contradictions, sparse support, lexical drift, and unsafe advice.

Main takeaway

TrustStack responds in the correct direction: stronger evidence raises the score and weaker evidence lowers it.

Best case

76.44 / 100

Worst case

59.97 / 100

Main bottleneck

Grounding + retrieval

Strong | Mixed | Weak

7. External Benchmark Findings

Public benchmarks test whether the same evaluation logic still works outside our own synthetic setup.

What this demonstrates

The system now runs on public tasks, not only internal tests.

Interpretation

Traceability is already strong. Retrieval and answer quality remain the next clear optimization target.

Aggregate TrustStack score

63.09 / 100

Task metric

0.141

Citation alignment

83.8%

8. Validity and Next-Stage Readiness

The current version is a strong baseline: measurable, reproducible, and ready for stronger next-stage validation.

What is already strong

- Reproducible local runtime
- Explicit scoring and risk logic
- Real benchmark coverage

What comes next

- stronger retrieval models
- richer human-study evaluation
- broader public benchmark coverage

Baseline → Benchmarks → Human studies

9. Conclusion

TrustStack makes AI trust easier to inspect, measure, and report.

Main contribution

It connects live evidence review, formal scoring, benchmark evaluation, and report-ready artifacts into one coherent workflow.

System role	Empirical basis	Deliverable
Evidence-first evaluator	**Synthetic + external**	**Report-ready artifacts**

Questions